Qian et al. 2020 — Breast Cancer

Breast · 10x (UMI counts) · 4,818 malignant cells · 7 samples = 7 patients · embryonic-program cell-state analysis

GO / NO-GO GATES

Gate 1 — NO-GO
mixing + continuity

Malignant cells form a per-patient archipelago. A single connected kNN graph appears only at the most generous integration (θ=10), where iLISI is just 1.1 of 7 samples — the scalar a naïve test reads as "integrated" — yet the embedding is organised BY PATIENT: 6/7 patients self-cluster far above chance (self-neighbour enrichment 6.8–23.5×, all past the 2.5× line). θ≤5 splits the manifold into two disconnected components, one PURE sample 54 (the dominant 65% tumour). Integration does not fix it — enrichment GROWS with θ and is PC-count-invariant (22× at n_pcs 20/30/40), so Harmony over-corrects. Every patient carries the SAME embryonic modules → inter-patient CNV/clonal heterogeneity, not a shared cell-state continuum → do not force a trajectory.

Gate 2 — N/A
module geography

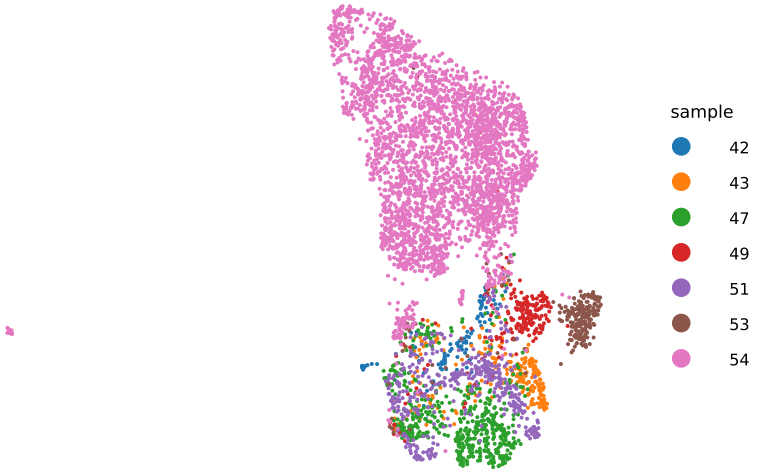
Not reached. Gate 2 tests a Gast/pSom hub and Crest geography on a fitted manifold; with Gate 1 NO-GO no manifold/trajectory is fit, so this gate does not apply.

Gate 3 — N/A
cross-repression

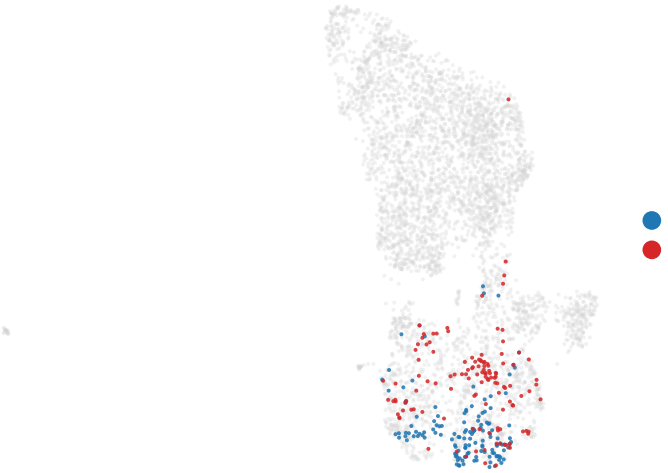
Not reached (branching-only, downstream of Gate 2). No hub, no bifurcation, no hub r.

STEP FIGURES

Malignant manifold — samples



EMT-TF expression — SNAI2 · FOXC1



PARAMETER CHOICES & RATIONALE

- **Sample threshold** ≥ 100 cells (protocol floor). Argmax multiplicative gap (3111→518, 6.01×) is the single dominant tumour above six well-populated samples, not a spillover tail near the floor → no clean count gap; floor keeps all 7/7 (min per-sample 105).
- **Gene / PCA** 1,225 overdispersed genes (scFates GAM); n_pcs = 20 (precedent-bounded — PC1 alone ≈ 21% variance, a dominant inter-patient / CNV axis; low scree elbow rejected per the evaluation audit).
- **Integration** Harmony on `sample`, θ swept 2 / 5 / 10 / 20 / 40. θ=10 = first single connected graph (most generous case); no θ mixes the patients (iLISI stays ≈ 1.1–1.4 of 7) — the choice is diagnostic, not corrective.
- **Root / geometry** None — no trajectory fit (Gate 1 NO-GO). Geometry: per-patient archipelago (no integrated continuum).
- **EMT-TFs shown** SNAI2 (11.3%) · FOXC1 (10.4%) — the two broadest detectable factors of a widened panel (FOXC / PRRX families added; VIM excluded as a marker, not a TF); overlaid without pseudotime (no trajectory).

VERDICT

The embryonic-program pattern does NOT reproduce in Qian 2020 breast cancer: the malignant compartment is a per-patient archipelago — no sample-integrated, continuous cell-state manifold. One patient (sample 54, 65% of cells) even forms a fully disconnected component at moderate integration. Gate 1 NO-GO — analysis stops before the trajectory (Gates 2 & 3 N/A). EMT-TFs SNAI2 and FOXC1 are the broadest detectable factors, reported over the un-integrated manifold.